

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 13



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Recap of Previous Lecture



Topic

Time & Distance



Topics to be Covered



Topic-1

Problem on Trains ✓

Topic-2

Boats and Streams ✓

Topic-3

Races ✓

[MCQ]



#Q. If a car travels along four sides of a square at 100 kmph, 200 kmph, 300 kmph and 400 kmph. Find its average speed.

Assignment

A

152 km/hr

B

180 km/hr

C

192 km/hr

D

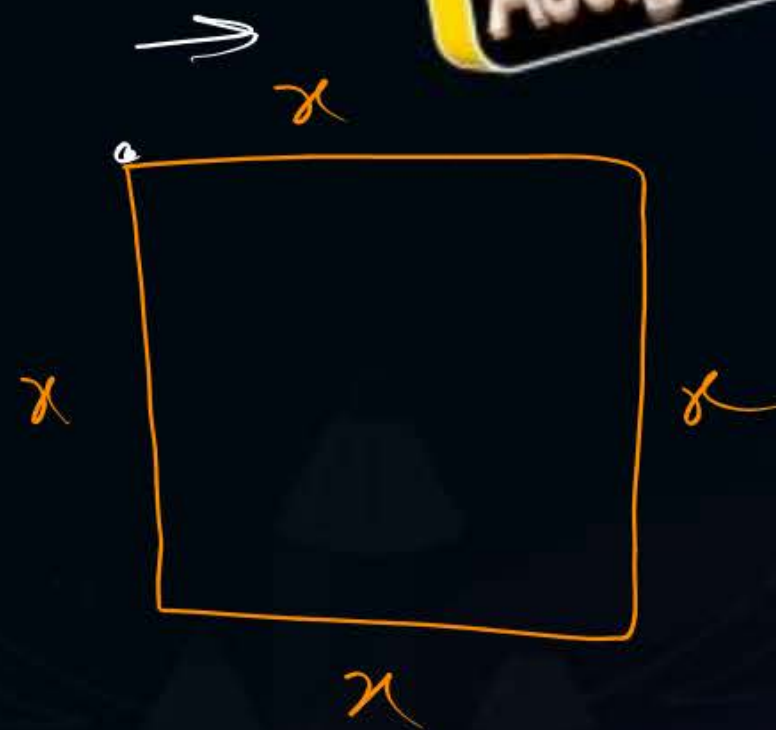
210 km/hr

$$\frac{4x}{A.S.} = \frac{x}{100} + \frac{x}{200} + \frac{x}{300} + \frac{x}{400}$$

$$\Rightarrow \frac{4x}{A.S.} = \frac{12x + 6x + 4x + 3x}{1200}$$

$$\Rightarrow \frac{4x}{A.S.} = \frac{25x}{300}$$

48



$$\therefore A.S. = 48 \times 4 = \underline{192 \text{ km/hr}}$$

[MCQ]

$$\frac{2 \times \frac{180 \times 18}{75}}{2} = 24 \text{ km/hr}$$



#Q. A tourist covers half of his journey by train at 60 km/hr, half of the remainder by bus at 30 km/hr and the rest by cycle at 10 km/hr. Average speed of the tourist during the journey is

- A 36 km/hr
- B 33 km/hr
- C 24 km/hr**
- D 18 km/hr

Assignment

Diagram illustrating the journey segments and speeds:

- Segment 1: Distance $2x$, Speed 60
- Segment 2: Distance x , Speed 30
- Segment 3: Distance x , Speed 10

Handwritten calculations:

$$\frac{2x}{60} + \frac{x}{30} + \frac{x}{10} = \frac{4x}{A.S.}$$
$$\Rightarrow \frac{2x + 2x + 6x}{60} = \frac{4x}{A.S.}$$
$$\frac{10x}{60} = \frac{4x}{A.S.}$$
$$\therefore A.S. = 24 \text{ km/hr}$$

Additional calculations shown:

$$\frac{2 \times \frac{180 \times 18}{75}}{2} = 24$$
$$\frac{2 \times 30 \times 15}{40} = 15$$

[MCQ]



#Q. Walking $\frac{3}{4}$ th of the usual speed a man is late to office by 10 minutes, find the usual time.

Assignment

$$\frac{4}{3} U \cdot T - U \cdot T = 10 \text{ min}$$

$$\left(\frac{4}{3} - 1\right) U \cdot T = 10 \text{ min}$$

$$\frac{1}{3} U \cdot T = 10$$

$$U \cdot T = 10 \times 3 = \underline{30 \text{ minutes}}$$

A

40 minutes

B

12 minutes

C

60 minutes

D

30 minutes

[MCQ]



#Q. Mumbai police observes a thief 800 metres away from him. The thief started running at 80 km/hr & the police at 170 km/hr. In how much time the police can catch the thief?

Assignment

- A** 8.88 minutes
- B** 8.88 hours
- C** 32 seconds
- D** 32 minutes

$$D = 800 \text{ m}$$

$$R.S. = 90 \text{ km/hr} \times \frac{5}{18} = 25 \text{ m/sec}$$

$$T = \frac{800 \text{ m}}{25 \text{ m/sec}} = 32 \text{ seconds}$$

[MCQ]



#Q. An employee goes to his office from his house at a speed of 60 km/hr and reaches his office 10 minutes late. If he follows the speed of 80 km/hr, he reaches his office 5 minutes early. Find the distance he covers to reach his office.

- A 55 km
- B 45 km
- C 50 km
- D 60 km**

$$\frac{x}{60} - \frac{x}{80} = \frac{15}{60}$$

$$T = \frac{D}{S}$$

$$D = 60 \text{ km}$$

$$D = S \times T$$

$$60(T + 10) = 80(T - 5)$$

$$T = \frac{S}{6}$$

$$\frac{x}{60} = \frac{1}{4}, \therefore x = 60$$

Assignment

10 am

60 kmph $\rightarrow$ 10:10
80 kmph $\rightarrow$ 9:55

[MCQ]

$$\frac{60}{144} = \frac{3}{2} = 1 \text{ hr } 30 \text{ min}$$



#Q. A man misses a train by 40 minutes if he travels at 30 kmph. If he travels at 40 kmph, then also he misses the train by 10 minutes. What is the minimum speed required to catch the train on time?

A 50 km/hr

B 55 km/hr

C 60 km/hr

D 45 km/hr

$$\frac{x}{30} - \frac{x}{40} = \frac{30}{60}$$

$$\Rightarrow \frac{x}{120} = \frac{1}{2}$$

D $\therefore x = 60$

$$T = \frac{60}{30} = 2 \text{ hrs}$$

$$A \cdot T = 1 \text{ hr } 20 \text{ min}$$

$$= 1 \frac{20}{60}$$

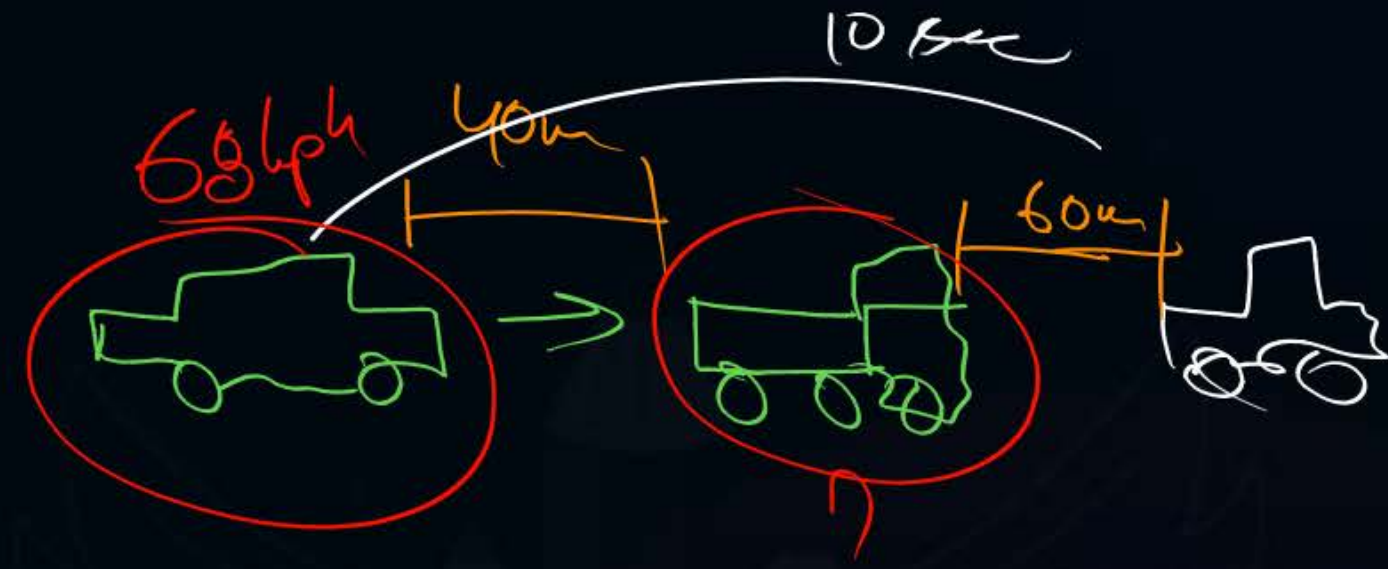
$$= \frac{4}{3} \text{ hr}$$

$$S = 60 \times \frac{3}{4} = 45 \text{ km/hr}$$

[MCQ]



#Q. A car driver driving at a speed of 68 km/hr [✓] locates a lorry 40 meters ahead of him. After 10 seconds, the lorry is 60 meters behind him. Find the speed of the lorry.



[MCQ]



#Q. A car driver driving at a speed of 68 km/hr locates a lorry 40 meters ahead of him. After 10 seconds, the lorry is 60 meters behind him. Find the speed of the lorry.

$$D = S \times T$$

$$D = 100m$$

$$R.S. = (68 - x) \text{ km/hr}$$

$$T = 10 \text{ sec}$$

$$R.S. = \frac{100}{10}$$

$$= 10 \text{ m/sec}$$

$$10 \times \frac{18}{5} = 68 - x$$

$$36 = 68 - x$$

$$\therefore x = 68 - 36$$

$$= 32 \text{ km/hr}$$



60m

40m

68 km/hr

10 sec

[MCQ]



#Q. Two boys begin together to write out a booklet containing 817 lines. The first boy starts with first line, writing at the rate of 200 lines an hour and the second boy starts with last line. He writes line 817 and so on backwards proceeding at the rate of 150 lines an hour. At what line will they meet?

A

469th

B

467th

C

465th

D

463rd

$$D = 817 \text{ lines}$$

$$R.S. = 350 \text{ lines/hr}$$

$$T = \frac{817 \text{ lines}}{350}$$

$$D = \frac{817}{350} \times 200$$
$$= \frac{3268}{7} = 466.86$$
$$\underline{467^{\text{th}}}$$

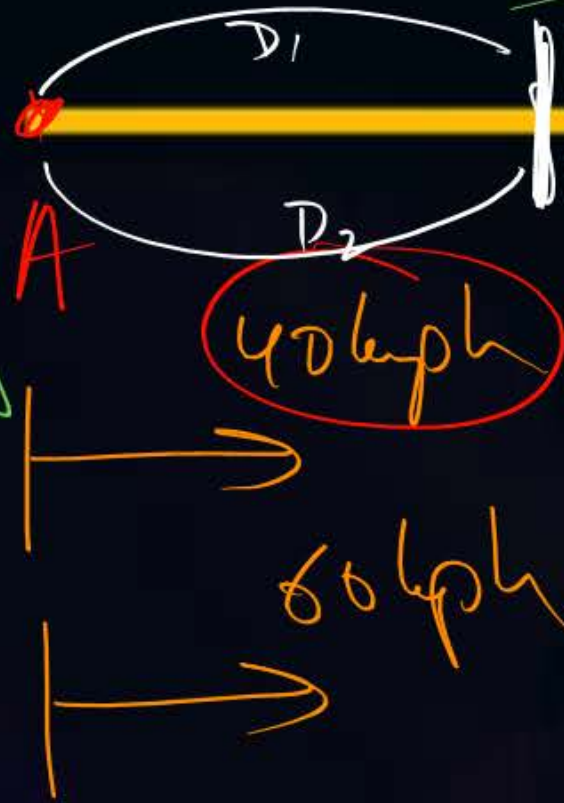
Meeting at a point:

$$D_1 + D_2 = T \cdot D \quad \text{opp.}$$

$\checkmark D = 600 \text{ km}$
 $\checkmark R.S. = 100 \text{ km/h}$
 $T = \frac{600}{100} = \underline{\underline{6 \text{ hrs}}}$

$$D_1 = D_2$$

Same



1200
 1800

600 km



$D = 600 \text{ km}$
 $R.S. = 20 \text{ kmph}$
 $T = \frac{600}{20} = \underline{\underline{30 \text{ hrs}}}$

Meeting at a point:

Two objects moving in opposite directions towards each other:

Distance covered by 1st object + Distance covered by 2nd object = Total distance



Two objects moving in same directions :

Distance covered by 1st object = Distance covered by 2nd object



[MCQ]



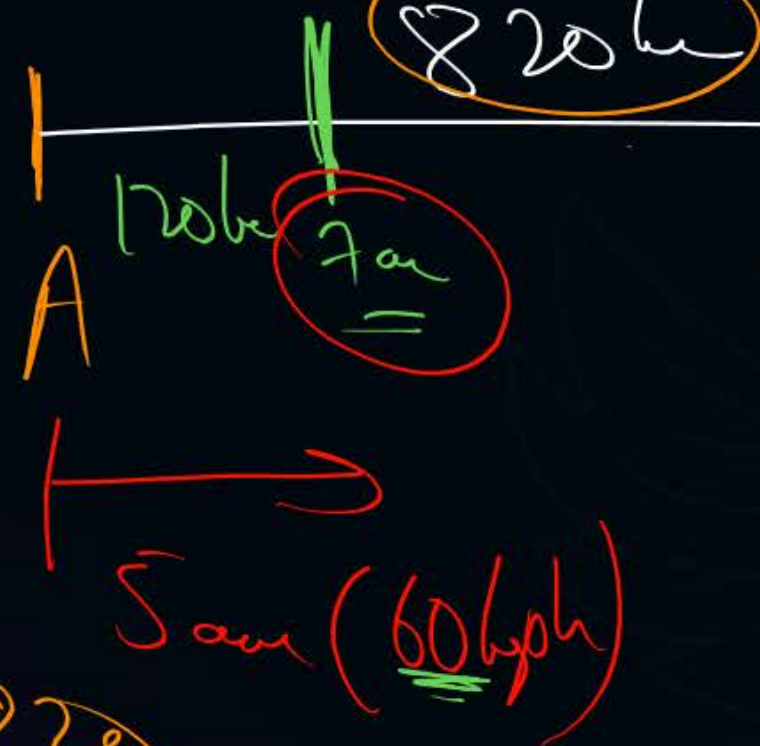
#Q. A train starts from Ahmedabad to Howrah at 5am everyday with the speed of 60kmph. Another train starts from Howrah to Ahmedabad at 7am everyday with the speed of 80 kmph. If the distance between Ahmedabad and Howrah is given as 820km, then at what time the two trains meet each other?

After 5am $\rightarrow$ 'x' hrs

$$820 = 60x + 80(x-2)$$

$$\Rightarrow 60x + 80x - 160 = 820$$

$$\Rightarrow 140x = 980$$



$$\therefore x = \frac{980}{140} = 7 \text{ hrs}$$

[MCQ]



$$\underline{D_1 = D_2}$$

#Q. A train starts from Chennai to Delhi at 11am with the speed of 110kmph.

Another train starts from Chennai to Delhi at 9am the same day with the speed of 90 kmph. If the distance between Chennai and Delhi is given as

1920km, then at what time the two trains meet each other?

$D = 180 \text{ km}$
 $R.S. = 20 \text{ kmph}$
 $T = \frac{180}{20}$

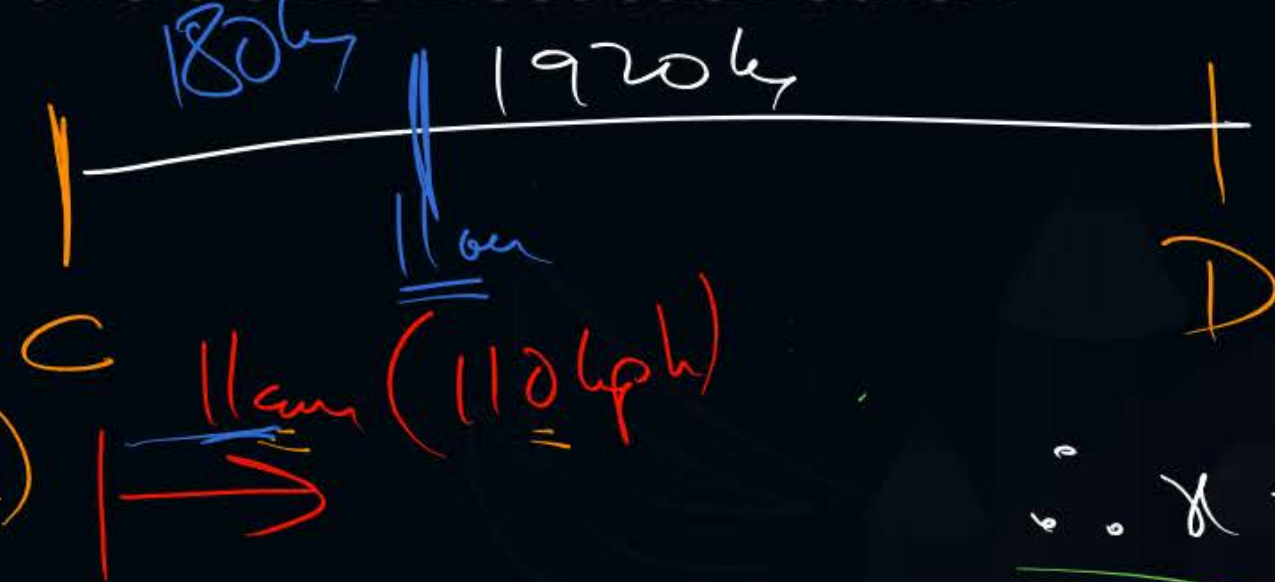
$= 9 \text{ hrs}$

After 9am $\rightarrow x$ hrs

$$90x = 110(x-2)$$

$$\Rightarrow 90x = 110x - 220$$

$$20x = 220$$



$$\therefore x = 11 \text{ hrs}$$

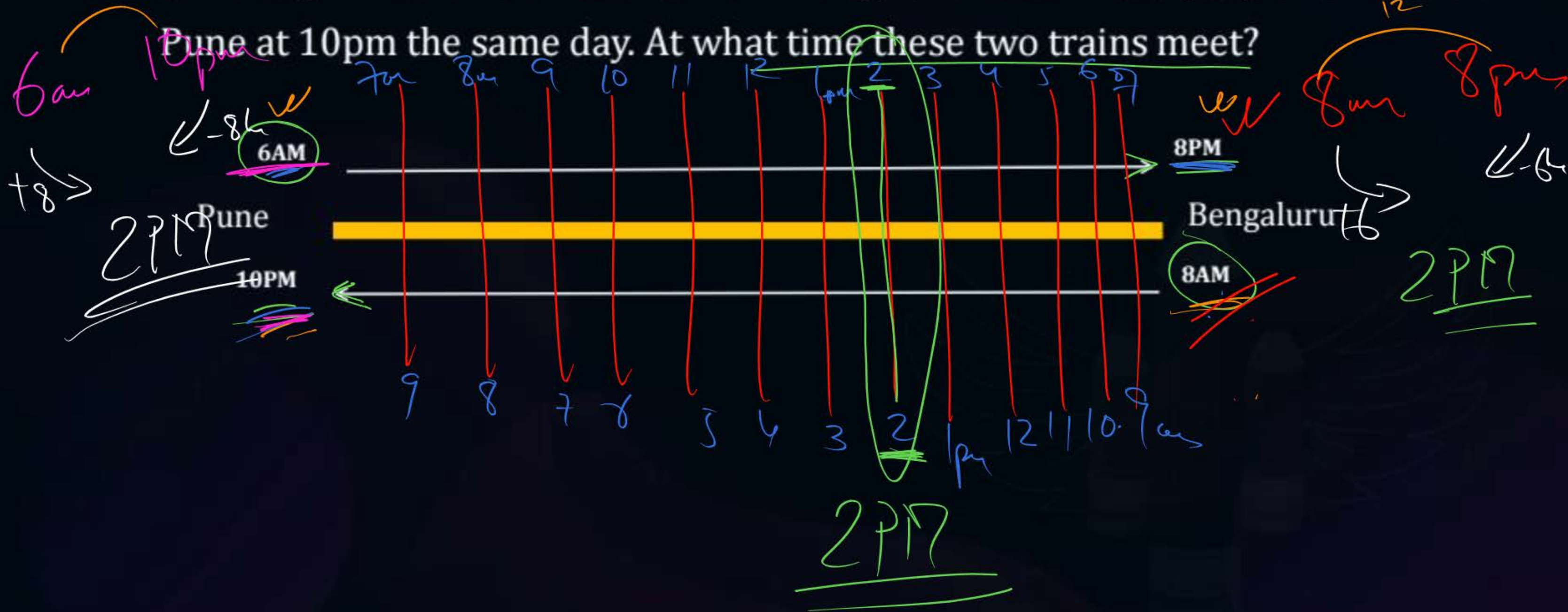
8 PM

[MCQ]

$$\frac{16}{2} = 8 \text{ hr}$$



#Q. A train starts from Pune at 6am everyday and reaches Bengaluru 8pm the same day. Another train starts from Bengaluru at 8am everyday and reaches Pune at 10pm the same day. At what time these two trains meet?



[MCQ]



#Q. A train starts from Leh at 5am everyday and reaches Kanyakumari at 9am the same day. Another train starts from Kanyakumari at 8am everyday and reaches Leh at 2pm the same day. At what time these two trains meet?

8:36 am

$$\frac{3}{8} \times 60 = 22.5$$



$$D_1 + D_2 = T.D.$$



$$8.36 \text{ am}$$

$$S_{T_1} = \frac{x}{4} \text{ mph}$$

$$S_{T_2} = \frac{x}{6} \text{ mph}$$

After 5 am $\rightarrow$ 'y' hrs

$$\left(\frac{x}{4} y\right) + \frac{x}{6} (y-3) = x$$

$$\Rightarrow \frac{y}{4} + \frac{y-3}{6} = 1$$

$$\Rightarrow 3y + 2y - 6 = 12$$

$$\Rightarrow 5y = 18$$

$$\therefore y = \frac{18}{5} = 3.6 \text{ hrs}$$

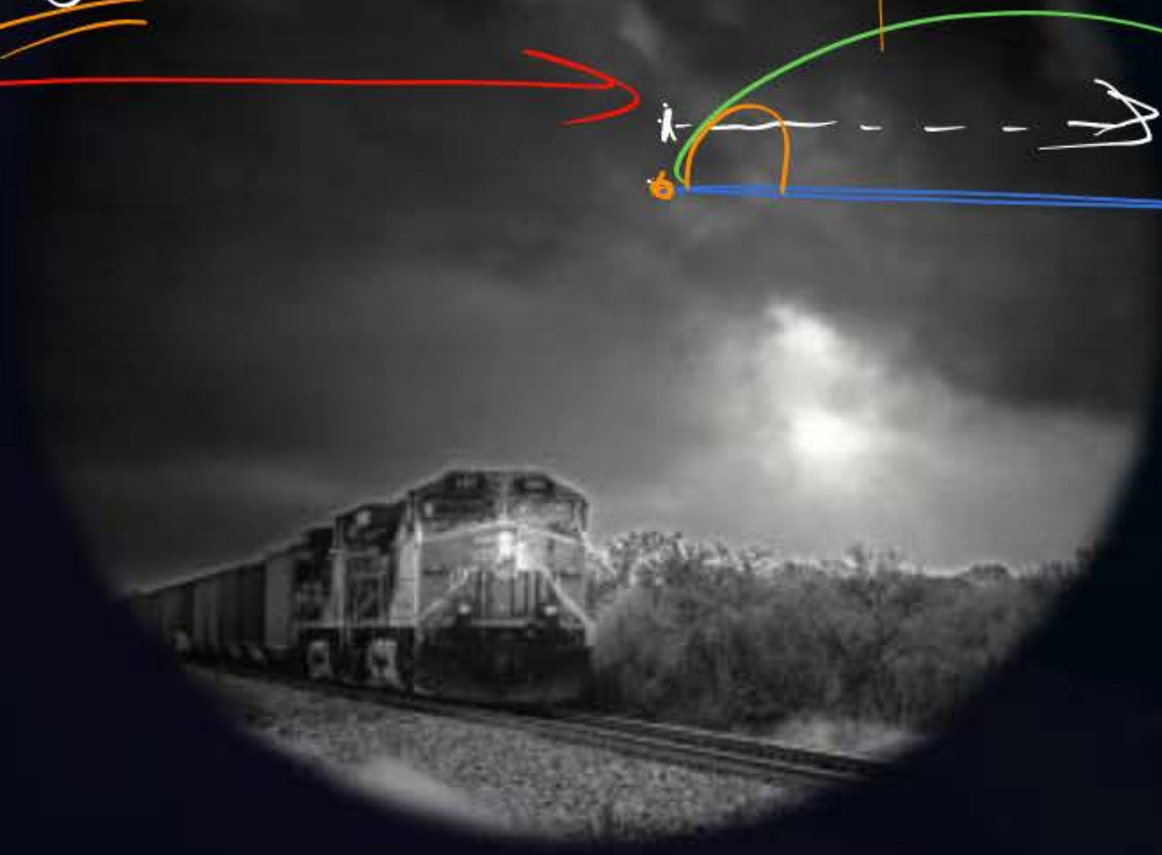
Trains



$$D = l_{\text{of Train}}$$

$$D = \underbrace{l_{\text{of T}}}_{\text{Train}} + \underbrace{l_{\text{of Platform}}}_{\text{Platform}} + \underbrace{l_{\text{of Bridge}}}_{\text{Bridge}} + \underbrace{l_{\text{of Tunnel}}}_{\text{Tunnel}}$$

The diagram illustrates the concept of distance D for a train crossing different structures. It shows a horizontal line representing the ground. A green arrow points from the left towards two stick figures standing on the ground. Below this, a red arrow points from the left towards a blue line representing a bridge. A green arc above the blue line represents the length of the bridge. A dashed line extends from the end of the bridge to the right. A red arrow points from the end of the dashed line towards the right. A blue line below the bridge represents the length of a tunnel. A dashed line extends from the end of the tunnel to the right. A red arrow points from the end of the dashed line towards the right.



Trains

$$D = l_{T_1} + l_{T_2}$$

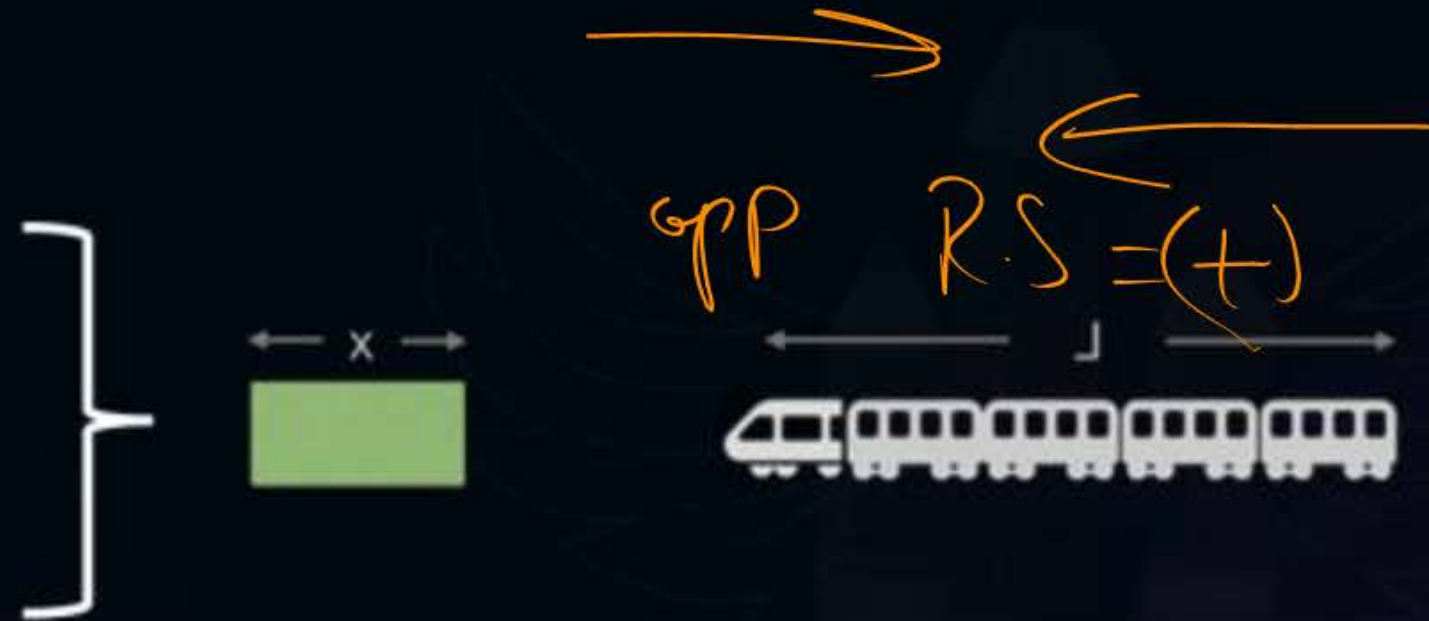
Same

$$R.S. = (\sim)$$

When a train try to cross a signal pole or standing man completely...
 D = length of the train



When a train try to cross a Platform or Bridge or Tunnel.....
 D = length of the train + length of Platform or Bridge or Tunnel



Trains



When a train try to cross another moving train.....

SAME DIRECTION : Speed (-)
OPPOSITE DIRECTION : Speed (+)

Distance = Length of Train 1 + Length of Train 2



[MCQ]



#Q. If a train travelling at 40 km/hr crosses another train of length 100 meter travelling at 14 km/hr in opposite directions in 30 seconds, then find the length of the train.

Assignment

[MCQ]



#Q. A train running at 52 km/hr takes 36 seconds to pass a platform. Next it takes 24 seconds to cross a man walking at the platform with 10 km/hr in the same direction. Find the length of the platform.

Assignment



2 mins Summary



Topic

Trains, Boats & Races

THANK - YOU